MAHFOOD BALFAQEEH

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EDUCATION

York University – *Hons. B.Sc. in Computer Science (Dean's Honour Roll)* Sep 2022 – Dec 2027
Coursework: DSA, OOP, Software Design, Computer Organization, Embedded Systems, Applied ML, Computation Theory

EXPERIENCE

Software Developer Intern | IBM - Skills Network Sept 2025 – Present | *Markham, ON*

- Cut streaming costs by **˜$20,000/mo** by refactoring **React/TypeScript** state, resolution encoding, and video sourcing.
- Accelerated AI Classroom chat speed by **10x** by engineering a full-stack **React, NestJS, and Redis** caching architecture.
- Unified **40+ LLMs across 5 APIs** with automated failover by building a centralized backend routing layer in **NestJS**.
- Automated real-time lab debugging by building a context-tracking service that parses live user browser states.

AI Intern | Growth-Hacker May 2025 – Aug 2025 | *Montreal, QC*

- Automated SEO keyword ranking at scale by building a **Python** scraping pipeline integrated with **SEMrush/Ahrefs APIs**.
- Eliminated manual SEO page updates by developing a web app linking live analytics to automated content generation.

PROJECTS

Awar – Object Detection Pipeline | *Python, OpenCV, Tkinter, YOLOv8* GitHub Repository

- Streamlined dataset creation by building a custom **Tkinter** GUI to annotate webcam frames into **YOLO** formats.
- Eliminated manual training setup by automating data splitting, preprocessing, and model training via **Ultralytics**.
- Achieved low-latency live object detection by deploying a real-time **YOLOv8** webcam inference pipeline in **OpenCV**.

Camera Tracker with BeagleBone AI | *Python, OpenCV* GitHub Repository

- Enabled real-time physical object tracking by programming closed-loop webcam servo controls on **BeagleBone AI**.
- Boosted video processing by **3x FPS** and cut latency by **50%** by optimizing multithreaded frame capture in **Python**.

SKILLS

Programming: TypeScript, Python, C/C++, C#, JavaScript, Java, Bash, MATLAB, HTML/CSS, SQL, NoSQL
Frameworks: React, NestJS, Node.js, Flask, PyTorch, OpenCV, JavaFX, Vite, Tailwind CSS
Developer Tools: Git, GitHub, AWS, Docker, Kubernetes, CI/CD, Linux/Unix, Redis, JUnit